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EDUCATION

University of California San Diego

Bachelor of Science in Electrical Engineering

La Jolla, CA

Sep 2024 - June 2028

EXPERIENCE

Triton Droids - Vice President of Engineering

La Jolla, CA | Jan 2025 – Present

- Implemented reinforcement learning through velocity based rewards, and motion reference clips for a human-like locomotion gait. Performed real to sim to real through Mjrlabs simulator. Deployed the first locomotion policy for our first generation humanoid, 1.3m, 31kg, 20 DOF.
- Derived per joint PD gains, friction coefficients, and armature inertias from Robstride actuator specifications and validated actuator dynamics in Isaac Lab to improve simulation fidelity for locomotion and sim to real deployment.
- Lead software, electrical, and mechanical engineering teams within a 100-member robotics club, coordinating development milestones, cross subsystem integration, and technical documentation across two humanoid generations.
- Designed and integrated system level requirements spanning Jetson/ESP32 compute, CAN motor control, IMU sensing, power distribution, battery management, safety layers, and serviceability.

PROJECTS

OpenRECAP - Self Improving Vision Language Action Model

Python, Pytorch, VLAs, MuJoCo, LeRobot, Reinforcement Learning

- Developed an offline reinforcement learning attachment to SmoIVLA, using autonomous rollouts, value estimation, and advantage labeled trajectory fine tuning; replaced the original large VLM value model with a lightweight state-based MLP to run the full pipeline on a single GPU.
- Built an end to end MuJoCo environment for the SO-101 integrating teleoperated behavior cloning in simulation, along with autonomous rollouts, reward labeling, value estimation, and iterative policy fine tuning.
- Dense rewards improved success from 50% to 60% and increased success for the most difficult position from 20% to 51%.
- Took a closed sourced, compute expensive technique in offline reinforcement + VLAs, and turned it into an open source lightweight model with similar architecture.

Humanoid Robot Fighting - Motion Imitation Learning

Python, MuJoCo, MJLab, PPO, RSL-RL

- Built a motion processing pipeline for the EngineAI T800 humanoid, mapping NPY/BVH boxing and martial arts motion onto a floating base and 25 actuated joints, with 50 Hz interactive replay and 1080p rollout rendering.
- Converted 30 Hz mocap into 50 Hz full-body reference trajectories using forward kinematics, quaternion SLERP, kinematic tree mapping, and finite difference velocity estimation.
- Adapted MJLab's PPO tracking stack to the T800, tracking 14 body segments with position, orientation, and velocity rewards plus joint limit, action rate, and self collision penalties. Trained policies capable of reproducing punch/kick motions, qualifying in the top 32 for the first round of the Ultimate Robot Knockout League.

Vision Language Action Models for Manipulation

Python, LeRobot, Action Chunking Transformers

- Fine-tuned and compared $\pi 0.5$ (VLA), ACT, and SmoIVLA on cube stacking. $\pi 0.5$ was the only policy to complete the full task, achieving 10% success versus 0% for ACT and SmoIVLA under varied cube positions.
- Fine-tuned $\pi 0.5$ for 50K training steps and diagnosed real world deployment failures caused by camera pose, lighting, and teleoperator distribution shift, improving repeatability with rigid 3D printed camera mounts and controlled lighting.
- Trained across H100 GPUs using VESSL AI, Google Colab, and SDSC. Tracked experiments with Weights & Biases and published checkpoints to Hugging Face.

SKILLS

Languages & Tools: Python, C++, MATLAB, PyTorch, LeRobot, MuJoCo, Isaac Lab, HuggingFace, Git, Linux/Ubuntu, Onshape, EagleCAD

Models & Methods: Reinforcement learning, $\pi 0.5$, ACT, SmoIVLA, PPO, SAC, Flow Matching, Behavior Cloning, Teleoperation, CAD/Mechanical Design